Vision-Only Latent Diffusion Super-Resolution without T2I Pretraining

sr-diffusion contributors

Experiment snapshot through active dual-context Stage 2 scale-up

Abstract

This report summarizes a work-in-progress x4 super-resolution system trained directly in vision space, without importing a pretrained text-to-image diffusion backbone. The completed pipeline consists of a factor-4 VAE, deterministic LR-to-HR-latent pretraining, conditional latent diffusion, and condition-start fine-tuning on a 103k-image photo split. The latest public XL-stage result uses a 500M-class full inference path: a wider Stage 2 condition encoder trained with a stronger degradation curriculum and a deeper 469.6M-parameter diffusion U-Net fine-tuned with decoded edge/highpass losses. On the current `photo_v3_noise_mix` sampled validation setup, the Stage 4 XL edge-loss checkpoint improves over bicubic by +0.7195 dB PSNR. Subsequent role-split and gated-residual probes show that bounded residual parameterization can nearly preserve the Stage 2 condition output on mild validation, reaching 25.0445 PSNR versus 25.0449 for condition-only, but it still does not provide a net average gain. A direct residual diagnostic then showed that the recoverable gap is dominated by high-frequency residuals, and a small deterministic sparse-gate residual refiner improved mild val100 mean PSNR from 25.0449 to 25.1178 with 86/100 wins over condition-only. A subsequent detail-preserving curriculum adaptation produced the first sampled gated-residual Stage 4 result to beat condition-only on both mean PSNR and a clear majority of samples: 25.3406 PSNR and 71/100 wins versus 25.3103 for the condition baseline. A decoded-detail residual refiner v2 then became the public Colab default, reaching 25.6410 PSNR on the photo-detail validation mix. The latest Stage 2 redesign adds multiscale context and balanced high-quality data; its selected step-46000 checkpoint improves the previous condition encoder across all tested presets. A VGG-feature-supervised continuation completed with small metric gains but no meaningful visible fine-detail improvement, so it was not promoted.

1 Objective

The target task is single-image x4 restoration:

$$128 \times 128 \to 512 \times 512.$$

A later extension targets

$$192 \times 192 \to 768 \times 768.$$

The design goal is a practical latent diffusion SR model trained from project checkpoints only. The system intentionally avoids Stable Diffusion, T2I backbones, text conditioning, and imported generative priors from pretrained diffusion models.

2 Architecture

The model is split into three trainable parts:

$$\text{HR image} \to \text{VAE} \to \text{HR latent},$$

$$\text{LR image} \to \text{condition encoder} \to \text{condition latent},$$

$$\text{noisy HR latent}, \text{condition latent}, t, \text{domain id} \to \text{conditional U-Net}.$$

The VAE decoder maps the denoised latent back to the 512px HR output. Domain IDs are implemented for photo and anime/illustration data, although the completed large-scale runs reported here use the photo split.

The stage numbers describe training order, not a mandatory serial inference chain. The public Colab default is

$$\text{LR} \rightarrow \text{Stage 2} \rightarrow \text{residual refiner} \rightarrow \text{Stage 1 decoder.}$$

For generative comparison, the runtime path is

$$\text{LR} \rightarrow \text{Stage 2} \rightarrow \text{Stage 3 or Stage 4} \rightarrow \text{Stage 1 decoder.}$$

Stage 4 is a Stage-3-derived replacement checkpoint, not a module applied after Stage 3.

3 Training Stages

The project is organized as staged training rather than a single end-to-end run:

1. Stage 1 trains the factor-4 VAE on HR crops.
2. Stage 2 trains a deterministic condition encoder to predict HR latents from degraded LR inputs.
3. Stage 3 trains a conditional diffusion U-Net in latent space.
4. Stage 4 fine-tunes diffusion from the condition latent at low timesteps.
5. Stage 5 is planned few-step distillation. If implemented, it will replace the slower Stage 3/4 sampler rather than run after Stage 4.
6. XL variants widen and deepen the condition encoder or diffusion U-Net while using partial checkpoint initialization where tensor shapes match.

4 Completed Implementation

The repository currently includes:

- manifest-based dataset loading with explicit domain IDs,
- on-the-fly x4 degradation pipelines,
- a factor-4 AutoencoderKL with 16 latent channels,
- deterministic LR-to-HR-latent pretraining,
- conditional latent diffusion with timestep and domain conditioning,
- DDIM/img2img sampling for validation,
- condition-start Stage 4 fine-tuning,
- condition-only validation tooling for isolating Stage 2 from diffusion,
- gated-residual $\hat{x}_0$ prediction for bounded Stage 4 refinement,
- Stage 2 residual oracle diagnostics for low/high-frequency error isolation,
- a deterministic bounded residual refiner with sparse/open gate ablations,
- bf16 CUDA/ROCm training paths,
- W&B logging, fixed validation samples, checkpoint evaluation, and recovery scripts for VM migration,
- partial checkpoint initialization for widened/deepened models,
- PyTorch DDP diffusion training with single-GPU fallback.

5 Data and Degradation

The large-scale photo runs use a 103,450-image training split and a fixed 100-image validation split. HR crops are 512px and LR inputs are 128px. The original photo degradation was extended with stronger blur, noise, compression, ringing, color shift, and banding variants. Earlier XL edge-loss work used `photo_v3_noise_mix`, a strong denoise-focused preset with no clean share and 80% combined v2/v3 cases. The latest `photo_detail_mix` uses 35% clean, 48% detail-preserving photo degradation, 15% mild degradation, and 2% strong `photo_v2`.

Item	Value
Task	x4 SR, 128px LR to 512px HR
Active large-scale domain	Photo
Train split	103,450 images
Validation split	100 fixed images
VAE latent	128px, 16 channels
Primary dtype	bf16
Latest detail degradation	`photo_detail_mix`

6 Completed Results

6.1 Stage 1 VAE

The selected Stage 1 VAE checkpoint is the best reconstruction artifact from the Stage 1 photo10k run. It was selected at the 50k-step checkpoint with:

Metric	Value
eval/recon	0.01198
eval/kl	9.38684
eval/psnr	40.19

6.2 Photo100k Baseline Stages

The first photo100k scale-up completed deterministic latent pretraining, conditional diffusion, and condition-start fine-tuning.

Stage	Run	Finished step	Selected result
Stage 2	photo100k b64 latent pretrain	30k	best latent loss 0.21230
Stage 3	photo100k b32 diffusion	60k	decoded PSNR 24.56 diagnostic
Stage 4	photo100k condition-start	5k	sampled val100 PSNR 25.4072

On the sampled validation comparison, the Stage 4 condition-start checkpoint slightly improved the Stage 3 checkpoint:

$$25.4072 - 25.3745 = +0.0327 \text{ PSNR}.$$

6.3 Photo100k v2 Denoise/Sharpening Stages

The `photo_v2` degradation curriculum produced lower absolute PSNR than the earlier degradation because the task is stronger, but Stage 4 improved both the Stage 3 v2 model and bicubic baseline on the sampled validation set.

Comparison	Sampled val100 PSNR
Bicubic baseline	22.4103
Stage 3 v2 SR	22.6699
Stage 4 v2 SR	22.8426

Stage 4 v2 improved over Stage 3 v2 by +0.1727 PSNR and beat bicubic by +0.4323 PSNR on the sampled validation set, with 81 wins and 19 losses in per-image comparison. Remaining artifacts included occasional color/contrast overshoot and small cyan/green sampling artifacts.

6.4 Stage 2 XL Condition Encoder

A smaller `photo_v3_noise_mix` Stage 2 run was stopped at step 12,700 after the validation latent loss plateaued near 0.282. The XL condition encoder then completed 80k steps with a larger 500M-class full-path target. Its main checkpoints were:

Checkpoint	Step	eval/latent_loss	eval/decoded_psnr
Best latent	66k	0.27230	21.38
Best PSNR proxy	72k	0.27940	21.52
Final/latest	80k	0.27593	21.51

The three Stage 2 XL candidates were then compared on the same 100 validation images by decoding condition-only outputs through the selected VAE. This comparison used fixed samples, fixed seed, and the same `latent_pretrain_photo100k_v3_noise_xl.yaml` configuration.

Candidate	Latent MSE	Mean decoded PSNR
Best latent, step 66k	0.91770	21.3828
Best PSNR proxy, step 72k	0.97295	21.5241
Latest, step 80k	0.89609	21.5062

The 72k-step checkpoint had the best decoded condition-only PSNR, so it was selected as the condition encoder for the next XL diffusion experiment.

6.5 Stage 4 XL Edge-Loss Condition-Start

The first Stage 4 XL run was then launched from the selected 72k-step Stage 2 XL condition encoder. The diffusion U-Net uses `base_channels=128`, `channel_multipliers=[1,2,4,8]`, six residual blocks per scale, and eight attention heads. Shape-compatible tensors were partially initialized from the smaller Stage 4 v2 checkpoint. The loss combines a small noise term, latent $\hat{x}_0$, decoded Charbonnier reconstruction, Sobel edge loss, and a highpass decoded loss.

Item	Value
Config	`diffusion_photo100k_xl_stage4_condition_v3_edge_b16.yaml`
Run	`diffusion_photo100k_xl_stage4_condition_v3_edge_b16`
Finished step	5000
Selected checkpoint	step 4250, best `eval/decoded_mse`
Training eval decoded PSNR	21.9872
W&B run	`https://wandb.ai/jwheo/sr-diffusion/runs/nog04fwr`

The selected checkpoint was sampled on the fixed 100-image validation split with condition initialization, 32 DDIM steps, and start timestep 50:

Metric	Value
Mean bicubic PSNR	22.3599
Mean SR PSNR	23.0793
Mean PSNR delta	+0.7195

Qualitatively, the XL Stage 4 checkpoint reduces noise and color contamination relative to the degraded LR/bicubic path. It is not yet a final restoration model: outputs remain visibly softer than the ground truth on fine textures.

6.6 Stage 4 XL Role-Split and Gated-Residual Probes

Follow-up probes tested whether Stage 4 could act as a bounded detail refiner instead of freely overwriting the deterministic Stage 2 condition latent. The role-split mild probe added a lowpass anchor and separated low/high-frequency decoded losses.

Item	Value
Config	role-split mild b8 probe
W&B run	`https://wandb.ai/jwheo/sr-diffusion/runs/lrb6nco9`
Finished step	8000 micro-steps, 2000 optimizer updates
Best one-step checkpoint	step 7500, decoded PSNR 23.2515

Sampled mild val100 showed that role-split losses reduced some over-editing at very low start timesteps, but the diffusion path still damaged the Stage 2 condition output on average:

Model	Start timestep	Mean SR PSNR	vs condition
Stage 2 condition-only	n/a	25.0449	n/a
Stage 4 role-split	25	24.5747	-0.4702
Stage 4 role-split	10	24.9185	-0.1264
Stage 4 role-split	5	24.9935	-0.0514
Stage 4 role-split	1	25.0335	-0.0114

The next probe changed the diffusion prediction type to `gated_residual_x0`. Instead of predicting the full denoised latent, the U-Net predicts a bounded residual and gate on top of the condition latent:

$$\hat{x}_0 = c + s \tanh(r)\sigma(g + b),$$

where c is the Stage 2 condition latent, r is the residual head, g is the gate head, $s = 1.25$, and $b = 0.0$. The probe was initialized from the role-split best checkpoint with partial initialization and stopped at step 2000 after sampled validation reached condition-only parity.

Item	Value
Config	gated-residual mild b8 probe
W&B run	`https://wandb.ai/jwheo/sr-diffusion/runs/edfko8e8`
Best one-step checkpoint	step 1000
Final checkpoint	step 2000

Model	Checkpoint	Start timestep	Mean SR PSNR	Wins vs condition
Stage 2 condition-only	n/a	n/a	25.0449	n/a
Stage 4 gated residual	1000	25	25.0415	25/100
Stage 4 gated residual	1000	10	25.0415	25/100
Stage 4 gated residual	1000	5	25.0416	25/100
Stage 4 gated residual	1000	1	25.0418	25/100
Stage 4 gated residual	2000	25	25.0445	34/100
Stage 4 gated residual	2000	10	25.0444	32/100
Stage 4 gated residual	2000	5	25.0444	31/100
Stage 4 gated residual	2000	1	25.0443	32/100

This is a partial success. Gated residual prediction almost eliminated the condition damage seen in previous Stage 4 probes and increased the number of samples beating condition-only from 10/100 at role-split $t = 1$ to 34/100 at gated-residual step 2000 with $t = 25$. However, the mean result still does not beat the Stage 2 condition-only baseline. The current interpretation is that Stage 4 has learned a safe near-identity residual path, not a reliable missing-detail generator.

6.7 Stage 2 Residual Diagnostic and Deterministic Refiner

A direct residual/oracle diagnostic was run on mild val100 to separate the Stage 2 condition-only error into lowpass and highpass components.

Metric	Value
Bicubic PSNR	24.4778
Condition decoded PSNR	25.0543
Oracle full residual PSNR	41.8207
Oracle full vs condition	+16.7664
Oracle highpass PSNR	35.0872
Oracle highpass vs condition	+10.0329
Oracle lowpass PSNR	25.0814
Oracle lowpass vs condition	+0.0270
Residual highpass energy ratio	0.8988
Residual lowpass energy ratio	0.0758

This indicates that the Stage 2 condition encoder already preserves most structure and low-frequency color, while the remaining recoverable error is dominated by high-frequency detail.

The follow-up deterministic bounded residual refiner freezes the Stage 1 VAE and Stage 2 condition encoder, then predicts only:

$$c + s \tanh(r)\sigma(g + b),$$

where c is the condition latent, r is the residual head, g is the gate head, s is the residual scale, and b is the gate bias. The sparse-gate probe reached its best validation result at step 500.

Model	Mean PSNR	vs condition	Wins	Gate mean
Stage 2 condition-only	25.0449	n/a	n/a	n/a
Sparse-gate residual refiner	25.1178	+0.0729	86/100	0.2147
Open-gate residual refiner	25.0972	+0.0523	73/100	0.8680

The sparse-gate result is a small but real gain and is qualitatively close to the condition output, without the destructive edits seen in previous diffusion Stage 4 probes. The open-gate ablation was worse despite a much larger mean gate value, so the next step should not simply force larger residual edits.

The sparse-gate refiner was then connected to standalone eval and inference scripts and evaluated without retraining across the active degradation presets.

Degradation	Bicubic	Condition	Refined	vs condition	Wins
`mild`	24.4778	25.0449	25.1178	+0.0729	86/100
`photo_v2`	22.4103	22.9271	22.9767	+0.0496	77/100
`photo_v3_noise_mix`	22.3599	22.9014	22.9600	+0.0586	86/100

Qualitatively, the refined outputs remain very close to the Stage 2 condition outputs. This is useful because the refiner does not introduce the destructive edits seen in earlier diffusion probes, but the visible detail gain is still small. The result is best interpreted as a safe residual teacher or warm start, not as a final detail generator.

7 Teacher-Supervised Stage 4 Residual Probe

The sparse-gate deterministic refiner was used as a frozen teacher for the gated-residual Stage 4 U-Net on `photo_v3_noise_mix`. Direct losses supervised the predicted residual, highpass residual, and gate. Training completed 8000 micro-steps, or 2000 optimizer updates with gradient accumulation four.

Checkpoint	Start t	SR PSNR	vs bicubic	vs condition	Wins
Teacher step 2000	25	22.9640	+0.6041	+0.0626	68/100
Teacher step 2000	50	22.9639	+0.6040	+0.0625	n/a
Teacher step 4000	25	22.9571	+0.5972	+0.0557	65/100
Teacher step 8000	25	22.9490	+0.5891	+0.0476	59/100
Edge step 4250	25	22.9563	+0.5964	+0.0549	42/100
Edge step 4250	50	23.0799	+0.7200	+0.1784	45/100

Teacher supervision produced a small cleanup gain over the Stage 2 condition output and slightly exceeded the edge model at $t = 25$. However, fur, leaves, branches, and building detail remained strongly smoothed. A simple absolute-Laplacian diagnostic measured teacher step-2000 output at 21.8% of GT detail energy, below the existing edge $t = 25$ output at 32.7%. This is not a complete perceptual metric, but it agrees with visual inspection.

The best sampled checkpoint was step 2000; continuing to step 8000 reduced both mean PSNR and condition win count. The current interpretation is that the `photo_v3_noise_mix` curriculum overemphasizes severe cleanup cases. The next high-signal experiment is therefore a less aggressive, detail-preserving degradation curriculum rather than another long continuation of this Stage 4 objective.

8 Detail-Preserving Curriculum Adaptation

A fixed-sample degradation audit confirmed that `photo_v3_noise_mix` overemphasized cleanup. On val100 it produced a 22.3599 bicubic PSNR baseline and 0.02040 mean LR chroma RMS error relative to clean downsampling. The new `photo_detail_mix` increased the bicubic baseline to 24.7357 and reduced chroma error to 0.00507.

The existing Stage 2 XL condition encoder already improved `photo_detail_mix` to 25.3103 PSNR, so it remained frozen. The teacher-supervised gated-residual Stage 4 was adapted for 12000 micro-steps with batch size eight, gradient accumulation four, and learning rate 10^{-6}.

Model/checkpoint	SR PSNR	vs bicubic	vs condition	Wins
Stage 2 condition-only	25.3103	+0.5745	n/a	n/a
Teacher Stage 4 initialization	25.3187	+0.5829	+0.0084	46/100
Photo-detail Stage 4 step 8000	25.3406	+0.6049	+0.0303	71/100
Photo-detail Stage 4 step 12000	25.3337	+0.5980	+0.0235	67/100
Existing edge Stage 4 step 4250	25.1176	+0.3818	-0.1927	13/100

Step 8000 is the selected checkpoint. It preserves the condition structure and sharpness and avoids the broad destructive edits produced by the edge model on this distribution. Mean absolute-Laplacian energy is still only 29.7% of GT, so the improvement comes primarily from more accurate bounded corrections rather than strong new texture synthesis. Rare strong-tail samples can still exhibit bright artifacts.

9 Decoded-Detail Residual Refiner v2

The deterministic residual refiner was expanded to 192 hidden channels and 12 residual blocks, with decoded-image and highpass supervision through the frozen Stage 1 decoder. A lower-learning-rate continuation completed 40,000 micro-steps, and step 39,000 was selected as the public default.

Degradation	Condition PSNR	Refined PSNR	Gain
`photo_detail_mix`	25.3103	25.6410	+0.3307
`mild`	25.0449	25.3161	+0.2712
`photo_v2`	22.9271	23.0419	+0.1148
`photo_v3_noise_mix`	22.9014	23.0787	+0.1773

This model is the default Colab path. It runs after the Stage 2 condition encoder and before the Stage 1 decoder; it does not use Stage 3 or Stage 4. Residual strengths 1.0, 0.75, and 0.5 expose an average-quality versus failure-tail trade-off.

10 Multiscale Stage 2 and Dual-Context Scale-up

The latest Stage 2 redesign preserves the selected XL predictor through partial initialization, then adds a zero-output-initialized multiscale context branch. Training exposure balances 100,000 COCO rows against 103,500 repeated DIV2K/Flickr2K rows. The 55.50M-parameter model completed 50,000 micro-steps, and step 46,000 was selected.

Degradation	PSNR gain vs previous Stage 2	Detail behavior
`photo_detail_mix`	+1.0348	slightly improved
`mild`	+0.9228	slightly improved
`photo_v2`	+0.9441	reduced on strong input
`photo_v3_noise_mix`	+0.9650	reduced on strong input

The selected model improves structure, color, and denoising but remains soft on fine texture. A continuation using frozen ImageNet VGG16 features completed 12,000 micro-steps from step 46,000. Step 8000 had the best shortlist score, 26.0092, and improved decoded PSNR over initialization by 0.0101 to 0.0256 dB across four presets. Step 11,000 was strongest on cleaner presets but regressed the strongest noise mix by 0.0063 dB. Fixed contact sheets showed almost no visible difference from initialization, and missing fine texture remained smoothed. The continuation was therefore not promoted.

The next scale-up changes capacity and data uniqueness together. The previous HQ-balanced manifest contained 203,600 rows but only 103,550 unique images. The new manifest adds 30,000 unique LSDIR training images, giving 133,450 unique training images plus the unchanged 100-image validation set. A second zero-output-initialized multiscale context branch preserves the selected step 46,000 output before training and increases predictor capacity from 55.50M to 119.24M parameters.

A one-L40S smoke test reproduced the initial 24.48 dB decoded PSNR and 0.291 detail ratio. Batch 8 with gradient accumulation 4 used approximately 37.8/46.1GB VRAM at 99% GPU utilization and approximately 0.75 micro-steps/s. The long run targets 100,000 micro-steps, or 25,000 optimizer updates. Validation remains every 1,000 micro-steps, while ordinary milestones are stored every 5,000 micro-steps to remain within disk budget.

11 Limitations

This is an engineering report for an active experiment, not a claim of a final published method. The large-scale completed results are photo-domain only, while anime/illustration domain conditioning remains implemented but not yet validated at the same scale. The reported sampled PSNR values are useful for internal checkpoint comparison, but they are not a full benchmark suite against external SR methods. The detail-preserving curriculum improved sampled Stage 4 accuracy, but the selected output remains conservative and does not recover all visible fine detail. The highest-signal next directions are a separate user-facing detail evaluation set, perceptual/detail metrics, and separation of the rare strong degradation tail into a robustness curriculum or evaluation slice. The completed VGG-feature continuation was not promoted into the public Colab default because its small metric gains did not translate into visible detail recovery.

References

[1] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Bjorn Ommer. High-Resolution Image Synthesis with Latent Diffusion Models. CVPR, 2022.

[2] Chitwan Saharia et al. Image Super-Resolution via Iterative Refinement. IEEE TPAMI, 2022.